MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

ICT4702 MODELING AND SIMULATION

END OF SEMESTER EXAMINATION

DATE: FEBRUARY, 2019 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

1. Answer **ALL** questions.
2. Marks for each question are indicated.
3. Write your answers clearly.
4. Indicate the question number for each answer..
5. This paper contains **FOUR** pages. Please check.

1.

a. Define the term simulation. **[2 marks]**

b. With the aid of a well labeled diagram, describe the steps involved in a simulation study. **[12 marks]**

c. In a simulation study, a modeler generally adopts a world view or an orientation for developing a model. Describe, any THREE typical world views you would adopt in developing a model. **[6 marks]**

d. **Figure 1** shows a simulation history (number of customers in the system against CLOCK time) of a single server FIFO system. Study the system and prepare a manual simulation table emphasizing customer number, arrival time, time service begins, time service ends and service time. **[5 marks]**

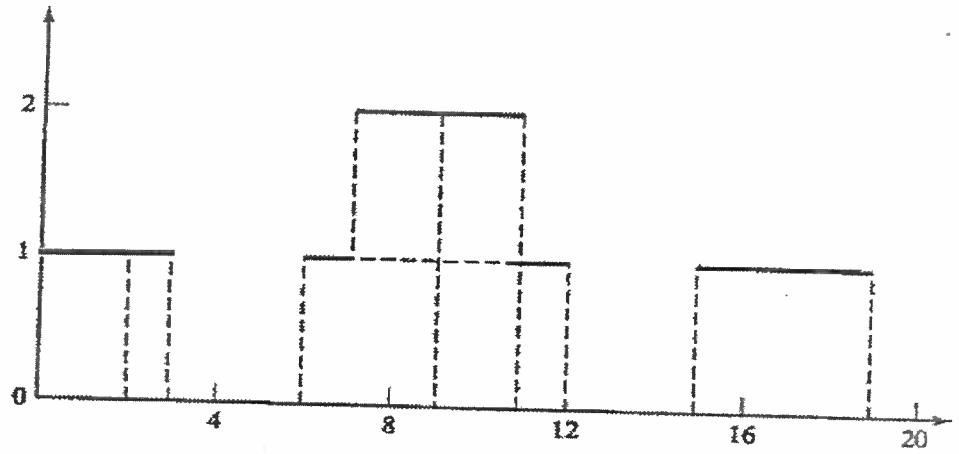


Figure 1

2. Students on **Zimba University** campus arrive at a library counter at random times that are from 1 to 8 minutes apart. Each possible value of the interarrival time has the same probability of occurrence. The time for a transaction at the counter is distributed as follows.

Transaction time (Minutes)	1	2	3	4	5	6
Probability	0.10	0.20	0.30	0.25	0.10	0.05

The following *(studentIdentity, interArrivalTimeRandomDigit, transactionTimeRandomDigit)* triples represent the first 7 students. The list is sorted by the *studentIdentity* attribute.

(1, - , 84); (2, 064, 18); (3, 112, 87); (4, 678, 81); (5, 289, 06); (6, 871, 91); (7, 583, 79);

a. For the 7 students find their

i. Interarrival times. **[3 marks]**

ii. Service times. **[3 marks]**

b. Simulate the system for the 7 students, making necessary assumptions, by

i. Process interaction method. **[7 marks]**

ii. Event scheduling method assuming system termination override at 30 minutes. **[8 marks]**

c. Which statistic(s) of the study in Question **2(b)** is observed to be sensitive to choice of the simulation method in this particular case? **[2 marks]**

d. How would you mitigate the artificial sensitivity referred to in Question **2(c)**? **[2 marks]**

3. Linear Congruential Method (LCM) produces a sequence of random integers, $X_1, X_2, \ldots$ by the following recursive relationship:

$$X_{i+1} = (aX_i + c) \bmod m, \quad i = 0, 1, 2, \ldots$$

a. Identify X_0, a, c and m. **[4 marks]**

b. Let $X_0 = 63$, $a = 19$, $c = 0$ and $m = 100$, find the next **FIVE** integers. **[5 marks]**

c. In some applications, it is useful to be able to quickly skip ahead in a pseudo-random number sequence without actually generating all the intermediate values. Also note the facts that for any integers g, h and m,

$(g+h) \bmod m = (g \bmod m) + (h \bmod m)$ and if $g = h \bmod m$ then $g = h - km$ for some integer $k \geq 0$. For an LCM with $c = 0$, show that

i. $X_{i+1} = (a^n X_i) \bmod m$. **[6 marks]**

ii. $(a^n X_i) \bmod m = (a^n \bmod m) X_i \bmod m$. **[6 marks]**

d. With the understanding developed in Question **3(c),** compute directly X_5 in Question **3(b).** **[4 marks]**

4.

a. The sequence of random numbers 0.54, 0.73, 0.98, 0.11 and 0.68 has been generated by an undocumented **georgeZimba.m** routine of a simulation package. Use the Kolmogorov-Smirnov test with $\alpha = 0.05$ and a critical value $D_{0.05} = 0.565$ to learn whether the hypothesis that the numbers are uniformly distributed on the interval [0,1] can be rejected. **[10 marks]**

b. Inverse Transform Technique is an algorithm for generating random variates for a desired probability distribution X given a sequence of random numbers R. The algorithm actually develops a random-variate generator $F^{-1}(X)$ as follows

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Algorithm: Inverse Transform Technique

Compute cumulative distribution function (cdf) of desired variable X
Set $F(X) = R$ on the range X
Solve the equation $F(X) = R$ for X in terms of R

Use the Algorithm to determine random-variate generators for the distributions with each of the following probability density function $f(x)$ or cdf's $F(X)$

i. $f(x)=\frac{x^2}{9}, \quad 0 \le x \le 3$ **[5 marks]**

ii. $$F(X)=\begin{cases} 1-e^{-\left(\frac{x}{\alpha}\right)^{\beta}}, & x \ge 0 \\ 0, & x<0 \end{cases}$$ **[4 marks]**

iii. $$F(X)=\begin{cases} 0, & x \le 0 \\ \frac{x^2}{2}, & 0<x \le 1 \\ 1-\frac{(2-x)^2}{2}, & 1<x \le 2 \\ 1, & x>2 \end{cases}$$ **[6 marks]**

END OF QUESTION PAPER